

```

pi@pi:~$ cd ~
pi@pi:~$ mkdir deepdream
mkdir: cannot create directory 'deepdream': File exists
pi@pi:~$ cd deepdream
pi@pi:~/deepdream$ git clone https://github.com/BVLC/caffe
fatal: destination path 'caffe' already exists and is not an empty directory.
pi@pi:~/deepdream$ cd ~
pi@pi:~$ sudo nano Makefile.config
pi@pi:~/deepdream$ cd ~
pi@pi:~/deepdream$ wget https://github.com/google/protobuf/releases/download/v2.6.1/protobuf-2.6.1.tar.gz
--2020-01-29 14:30:17-- https://github.com/google/protobuf/releases/download/v2.6.1/protobuf-2.6.1.tar.gz
Resolving github.com (github.com)... 13.234.176.102
Connecting to github.com (github.com):443... connected.
HTTP request sent, awaiting response... 301 Moved Permanently
Location: https://github.com/protocolbuffers/protobuf/releases/download/v2.6.1/protobuf-2.6.1.tar.gz [following]
--2020-01-29 14:30:18-- https://github.com/protocolbuffers/protobuf/releases/download/v2.6.1/protobuf-2.6.1.tar.gz
Reusing existing connection to github.com:443.
HTTP request sent, awaiting response... 302 Found
Location: https://github-production-release-asset-2e65be.s3.amazonaws.com/23357588/0a2433bc-5a29-11e4-8e74-fbea8721fcc7?X-Amz-Algorithm=AWS4-HMAC-SHA256&X-Amz-Credential=AKIAIWNJYAX4CSVEH53A%2F20200129%2Fus-east-1%2F%3F2Faws4_request&X-Amz-Date=20200129T103018Z&X-Amz-Expires=300&X-Amz-Signature=f78f56193bca4318b7cd6e3e95dd2d0879d072da74712f6ae6444970fc1164f6&X-Amz-SignedHeaders=host&actor_id=0&response-content-disposition=attachment%3B%20filename%3Dprotobuf-2.6.1.tar.gz&response-content-type=application%2Foctet-stream [following]
--2020-01-29 14:30:19-- https://github-production-release-asset-2e65be.s3.amazonaws.com/23357588/0a2433bc-5a29-11e4-8e74-fbea8721fcc7?X-Amz-Algorithm=AWS4-HMAC-SHA256&X-Amz-Credential=AKIAIWNJYAX4CSVEH53A%2F20200129%2Fus-east-1%2F%3F2Faws4_request&X-Amz-Date=20200129T103018Z&X-Amz-Expires=300&X-Amz-Signature=f78f56193bca4318b7cd6e3e95dd2d0879d072da74712f6ae6444970fc1164f6&X-Amz-SignedHeaders=host&actor_id=0&response-content-disposition=attachment%3B%20filename%3Dprotobuf-2.6.1.tar.gz&response-content-type=application%2Foctet-stream
Resolving github-production-release-asset-2e65be.s3.amazonaws.com (github-production-release-asset-2e65be.s3.amazonaws.com)... 52.216.92.155
Connecting to github-production-release-asset-2e65be.s3.amazonaws.com (github-production-release-asset-2e65be.s3.amazonaws.com):443... connect
ed.
HTTP request sent, awaiting response... 200 OK
Length: 2641426 (2.5M) [application/octet-stream]
Saving to: 'protobuf-2.6.1.tar.gz.1'

protobuf-2.6.1.tar.gz.1 100%[=====] 2.52M 522KB/s in 6.6s

2020-01-29 14:30:27 (389 KB/s) - 'protobuf-2.6.1.tar.gz.1' saved [2641426/2641426]

pi@pi:~/deepdream$

```

Fig. 6: Cloning Deedream

```

pi@pi:~/deepdream$ tar xf protobuf-2.6.1.tar.gz
pi@pi:~/deepdream$ cd protobuf-2.6.1
pi@pi:~/deepdream/protobuf-2.6.1$ sudo ./configure
checking whether to enable maintainer-specific portions of Makefiles... yes
checking build system type... armv7l-unknown-linux-gnueabi
checking host system type... armv7l-unknown-linux-gnueabi
checking target system type... armv7l-unknown-linux-gnueabi
checking for a BSD-compatible install... /usr/bin/install -c
checking whether build environment is sane... yes
checking for a thread-safe mkdir -p... /bin/mkdir -p
checking for gawk... no
checking for mawk... mawk
checking whether make sets $(MAKE)... yes
checking whether make supports nested variables... yes
checking for gcc... gcc
checking whether the C compiler works... yes
checking for C compiler default output file name... a.out
checking for suffix of executables...
checking whether we are cross compiling... no
checking for suffix of object files... o
checking whether we are using the GNU C compiler... yes
checking whether gcc accepts -g... yes
checking for gcc option to accept ISO C89... none needed
checking whether gcc understands -c and -o together... yes
checking for style of include used by make... GNU
checking dependency style of gcc... gcc3
checking for g++... g++
checking whether we are using the GNU C++ compiler... yes
checking whether g++ accepts -g... yes
checking dependency style of g++... gcc3
checking how to run the C++ preprocessor... g++ -E
checking for grep that handles long lines and -e... /bin/grep
checking for egrep... /bin/grep -E
checking for ANSI C header files... yes
checking for sys/types.h... yes
checking for sys/stat.h... yes
checking for stdlib.h... yes
checking for string.h... yes
checking for memory.h... yes
checking for strings.h... yes
checking for inttypes.h... yes

```

Fig. 7: Installing protobuf

```

libgtk-3-dev
sudo apt-get install libatlas-base-dev gfortran

```

After that swap the file size. To do so, run the following command:

```
sudo nano /etc/dphys-swapfile
```

Then change the line CONF_SWAPSIZE = 100 to CONF_SWAPSIZE = 1024

```
sudo /etc/init.d/dphys-swapfile stop
```

```

sudo /etc/init.d/dphys-swapfile start
sudo pip3 install opencv
sudo pip3 install numpy
wget https://bootstrap.pypa.io/get-pip.py
pip install dlib

```

Fig. 4 shows how to install dlib required for face recognition.

Next, to install the face recognition library, run the following command in the terminal:

```
sudo pip3 install face_recognition
```

Run the sudo apt-get update command again; with that the basic setup for emotion recognition is done.

For deep dream requirements/setup, install the following modules in Python:

- Caffe
- Protobuf (also known as Protocol Buffers)
- PyCam

Open your Linux terminal and run the above commands to install the modules (refer Fig. 5 through Fig. 11). You can also follow the Google Deep Dream official site for installation on Raspberry Pi.

Coding

Having completed the basic setup of the libraries, write the code for real-time face recognition. First, download OpenCV, OpenCV Contribution, and Face Recognition library folder. To do so, open the terminal window and run the following command in the terminal:

```

git clone https://github.com/opencv/opencv
git clone https://github.com/opencv/opencv_contrib

```